



CAPSTONE PROJECT · TRACK C

How well can we forecast next month's reservoir pressure using production and injection data?

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Machine Learning for Petroleum Engineers & Geoscientists · Final Project Presentation



Why pressure forecasting matters

THE QUESTION

Can we forecast reservoir pressure for next month from injection and production history?

Forecasting pressure is important to monitor pressure declines before they become a problem and force costly interventions



Who uses the answer

Reservoir engineers, production engineers



The decision it supports

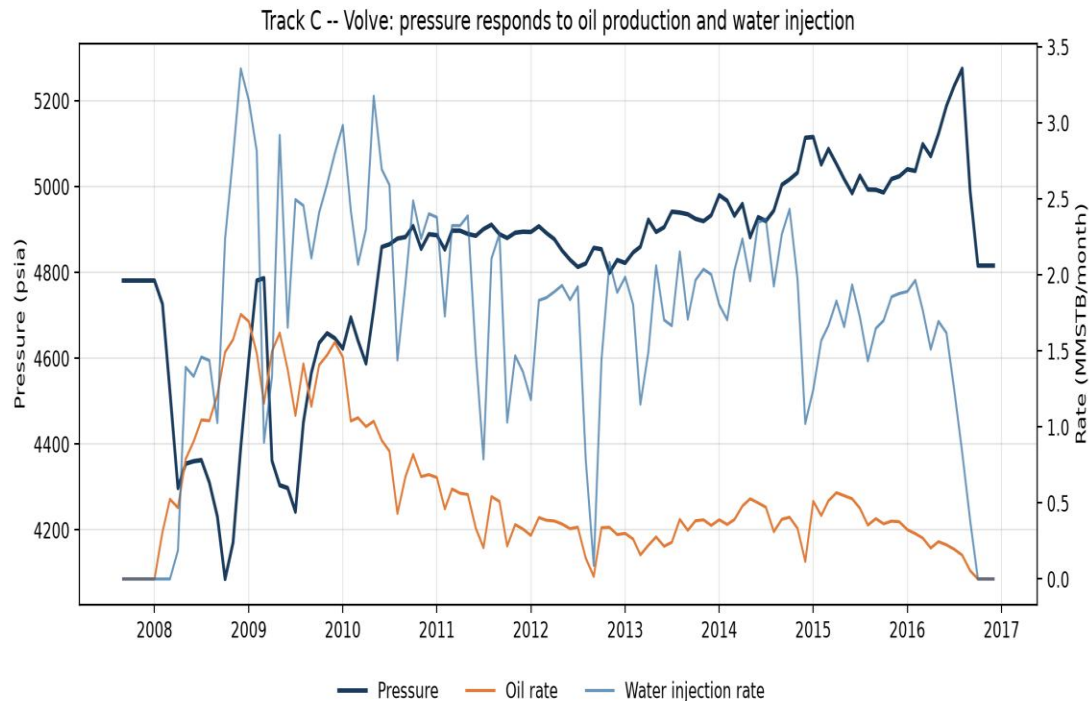
Deciding when to adjust water injection rates, by catching pressure declines early



Success criterion

A model that beats the persistence baseline

Volve: Pressure, Water Injection and Oil Production



Pressure drops with oil production, and recovers under water injection

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112

months

88

training months

3

input features

SOURCE & CLEANING

- Source: Volve field dataset, released publicly by Equinor (Equinor Volve data license)
- Missing values: No missing months in the 2007-09 – 2016-12 window used
- Transforms: Cumulative production and injection volumes converted to monthly rates via `.diff()`
- One known data-quality issue: Simulation-derived field averages, not gauge data; production starts in 2008-02.

Chronological Time Split

THE SPLIT- LAST 18 MONTHS HELD OUT

88 months
(2008-03-01 to 2015-06-01)

Train

18 months
(2015-07-01 to 2016-12-01)

Test (never seen)

last 18 months held out – This split matches real use because in real life historical data is used to forecast future values which have not yet been seen.



Metric

RMSE in pressure units, psia

MAE in pressure units, psia

RMSE penalises the large misses that trigger interventions; MAE shows typical error



Baselines

Naive: persistence

Classical: random forest



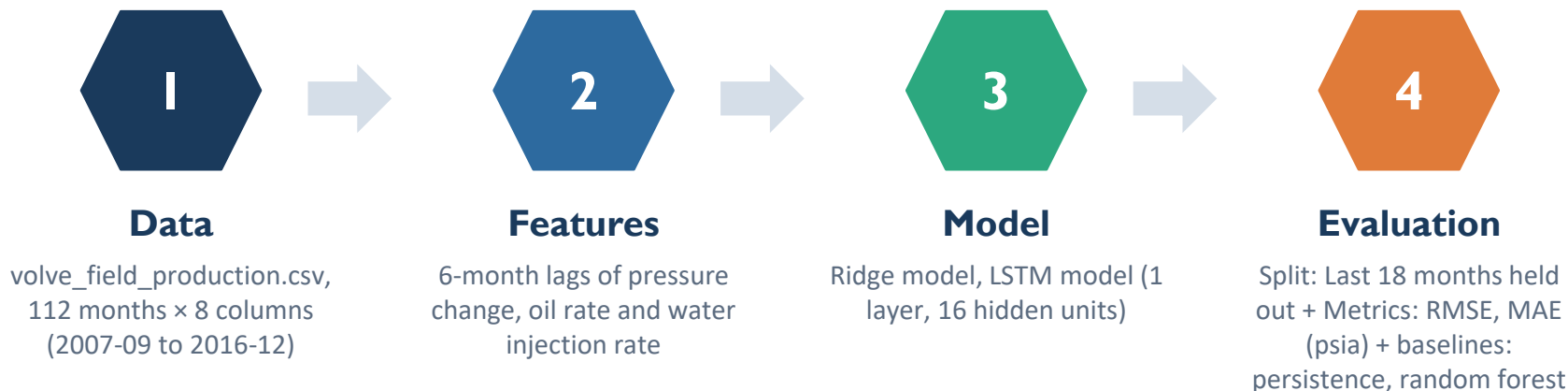
Leakage checked

Features excluded: GOR, Gas Injection Rate, Gas Production Rate

Scalers fitted on training data only



Project Pipeline



TUNING (TRAINING DATA ONLY)

Ridge strength and lag length (3, 6, 12 months) chosen by walk-forward CV (TimeSeriesSplit)
Scalers fitted on training months only

WHAT I TRIED THAT DIDN'T WORK

Forecasting oil rate instead of pressure: no model beat persistence
Adding drivers to the LSTM: no gain on the test months
6-month recursive forecasts: no better than a flat line

Results: Best model vs. Baseline

Model	RMSE (psia)	MAE (psia)	vs. best baseline
Naive baseline (Persistence)	85.3	49.5	--
Classical baseline (Random Forest)	95.1	68.4	-11.5%
LSTM	90.7	72.3	-6%
Ridge	82.0	50.3	4%

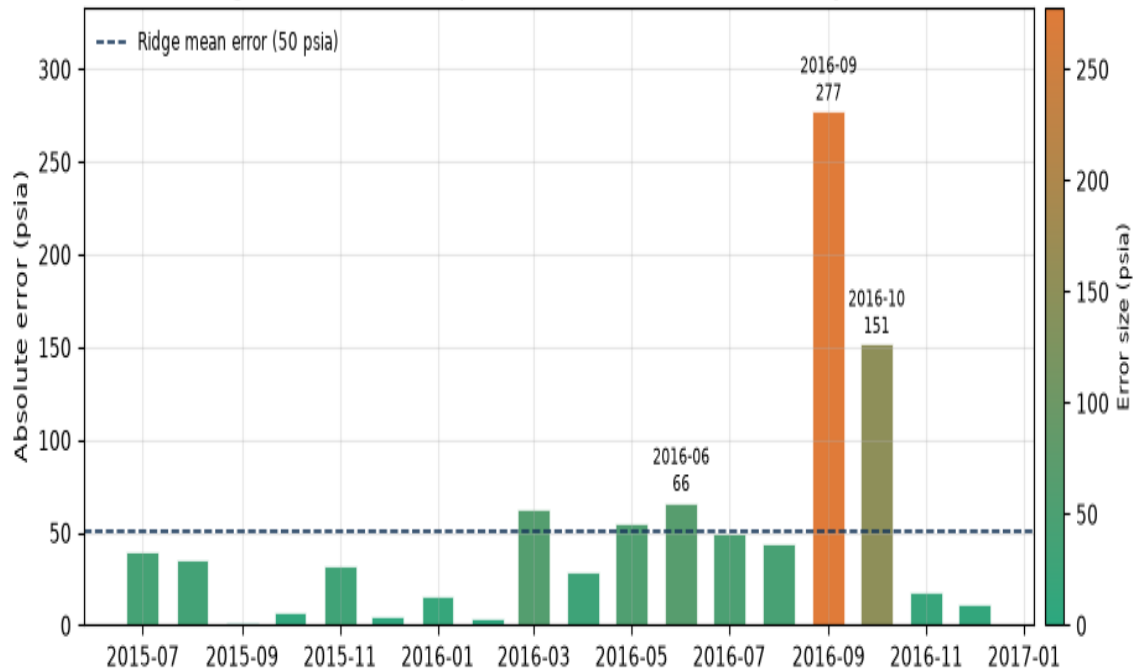
All rows evaluated on the same split: last 18 months held out. Green row = final model.

[+4%]

3.9 % lower RMSE; MAE not better on unseen test data

Where it fails: Sudden shut – in

Ridge on dP (LAGS=6) -- pressure + drivers: absolute error by month



Finding 1

Error was largest at shut in on 2016-09 and 2016-10. It was off by +277 psia and +151 psia



Finding 2

inputs show the slowdown but not the shut-in; training data contains no shut-in to learn from.



Finding 3

Adding a driver that indicates shut in could prove to be useful

How trustworthy is the Ridge Model?

HEADLINE UNCERTAINTY

± 70 psia

% prediction interval from walk-forward training errors

Checked on held-out data: Covers 16 of 18 held-out months (89%); 100% excluding the shut-in

USE IT FOR

- Month-ahead check of field-average pressure in stable operation, for Volve-type fields
- Benchmarking any fancier forecast against persistence

DON'T USE IT FOR

- Anticipating shut-ins or rate changes
- Forecasts beyond one month
- Wells or fields other than Volve
- Forecasting oil rate

Conclusions

- Ridge reaches RMSE 82.0 psia vs 85.3 for persistence on the 18 held-out months.
- The 3.9% gain is within noise: a tie, not a win.
- The 2016 shut-in causes most of the error, and no model anticipates it.



Limitations and next steps



Limitations

- 112 monthly points (88 for training, 18 for testing).
- Two shut-in months (2016-09, 2016-10) produce most of the test RMSE.
- Field-level data derived from a reservoir-simulation history match, not raw gauge data.



Next steps

- Add an operational flag (planned shut-in / rate-cut schedule)
- Evaluate with rolling-origin backtests over the whole history, not one 18-month window.
- Test well-level pressure data (Volve gauges) rather than a field average.

THE TAKEAWAY

The ridge model can forecast pressure after a month using oil production and water injection rates to within ± 70 psia, 90% of the time; but not reliably better than the persistence baseline

github.com/sadaraka-kadi/ml4pe-pressure-forecasting





Thank you -- questions?

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Backup: model card

Model name & version	volve-pressure-ridge-dP v1.0	Evaluation split	Chronological: last 18 months (2015-07 to 2016-12) held out
Intended use	Month-ahead sanity check on Volve-type field-average pressure for reservoir/production engineers	Headline performance	RMSE 82.0 psia, MAE 50.3 psia vs persistence 85.3 / 49.5 psia
Not intended for	Forecasting shut-ins or rate changes; horizons beyond 1 month	Where it fails	2016-09 and 2016-10 shut-in: errors +277 and +151 psia
Training data	88 training months 2008-03 to 2015-06	Uncertainty	Empirical 90% interval +/-70 psia
Inputs & units	Monthly pressure changes (psia); oil rate production rate and water injection rate (MMSTB/month)	Limitations & risks	Small sample; simulation-derived data; one test window; interval assumes no regime change
Output & units	Next-month pressure (psia)	How to reproduce	https://github.com/sadaraka-kadi/ml4pe-pressure-forecasting ; Restart & Run All